



FAKE NEWS DETECTION



A MINI PROJECT REPORT

Submitted by

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ABSTRACT

The proliferation of fake news on social media platforms and news websites has become a significant societal problem, threatening the credibility of information sources and undermining public trust. In response to this challenge, we present a project focused on the development of an automated fake news detection system. Leveraging the power of machine learning, our project aims to provide an effective solution for identifying and flagging misleading or fabricated news articles. Our preliminary results indicate promising accuracy and precision in detecting fake news articles. By harnessing the strengths of machine learning algorithms, our system exhibits the potential to play a crucial role in curbing the spread of false information. The project's success would aid in enhancing media literacy, preserving the integrity of journalism, and ultimately fostering a more informed and responsible society. This project involves three primary phases: data collection, feature engineering, and model development. We curate a comprehensive dataset containing both real and fake news articles, ensuring diversity in content and sources. Feature engineering involves text preprocessing, sentiment analysis, linguistic pattern recognition, and source credibility assessment. Machine learning models and traditional classifiers are trained and evaluated using the given dataset and it is detected as fake news or a real news with the model.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

. In an era dominated by the rapid dissemination of information through digital channels, the rise of fake news has emerged as a pervasive and challenging issue. The internet and social media platforms have revolutionized the way we consume news and information, but they have also provided a breeding ground for the spread of misinformation, disinformation, and fake news articles. These fabricated or misleading stories have the potential to deceive, manipulate, and sow discord, posing a serious threat to the credibility of information sources and undermining public trust in the media. The consequences of fake news are far-reaching. It can influence public opinion, distort facts, and even incite social and political unrest. As a result, the need for effective tools and strategies to combat the proliferation of fake news has never been more critical. This project is a response to this pressing need. We aim to leverage the power of machine learning and natural language processing (NLP) to develop an automated system for detecting and flagging fake news articles. Our project is guided by the belief that technology, when harnessed intelligently, can serve as a potent weapon against the onslaught of misinformation. In the following sections, we will delve into the intricacies of our approach, from data collection and preprocessing to feature engineering and model development. By the end of this project, we aspire to offer a robust solution that can help identify fake news, thereby contributing to media literacy and fostering a more informed and responsible society. The fight against fake news requires a multidisciplinary effort, and this project represents a significant step in that direction, combining the fields of data science, linguistics, and information technology. As we progress through this endeavor, we hope to make a meaningful contribution towards addressing the challenges posed by fake news in the digital age.

1.2 Objective for the project

- Fake News Detection has a main objective in prediction of the false information is a specific well or division in advance by using various regression techniques.
- This model also helps the society by identifying the misleading information helping to prevent the spread of fake information.

1.3 Problem Statement

To develop a machine learning system to automatically identify and classify fake news articles from a mixed news dataset, with the objective of reducing the spread of misinformation and enhancing public trust in online news sources.

1.4 Scope of the project

- Machine learning covers various classifiers of Supervised, Unsupervised and Ensemble Learning which are used to predict and find the accuracy of the given dataset.
- We can use that knowledge in our project of Fake news detection as it will help a lot of people.
- To establish a proper fake news detector, ensemble method is required. So that the prediction news can be more accurate and can't spread to all parts of the country

CHAPTER 2

ARCHITECTURE

2.1 Architecture Diagram

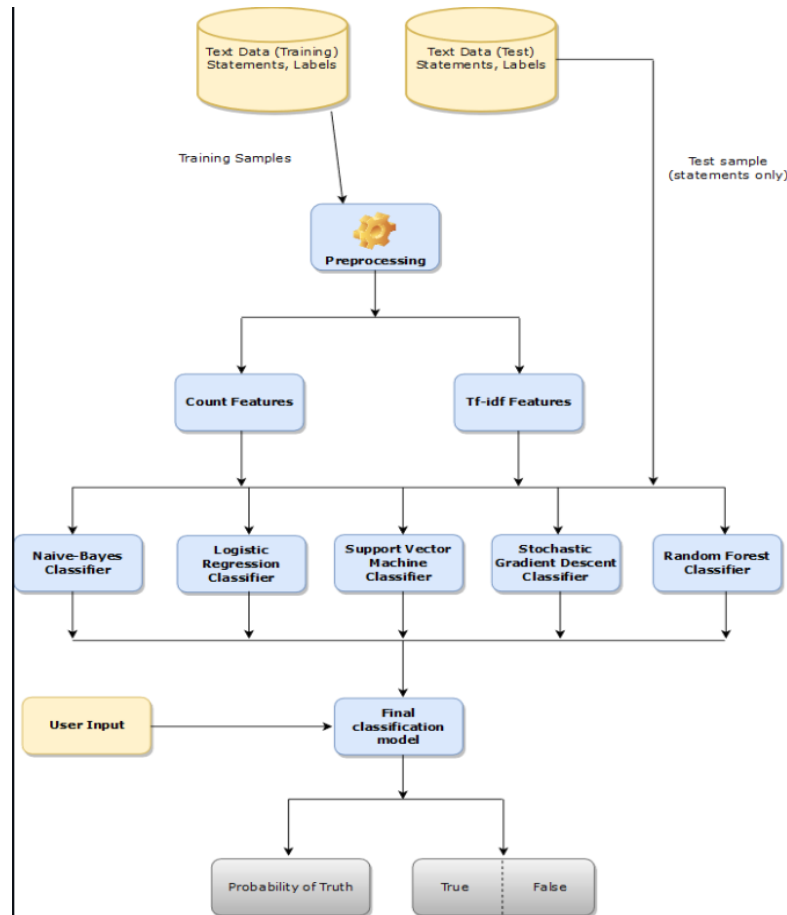


Figure 2.1.1 – Architecture of the model

This model is constructed by taking a Fake news dataset and identifying the variables. The Pre-processing of the dataset is carried out in order to make the required target and class variables. Now the Feature reduction is done and the data is split up into training data and the testing data to predict the values. After this splitting of the data these are passed to the various prediction models and the accuracy of the model is calculated and the results are displayed.

CHAPTER 3

Working

3.1 Literature Review

The system predicts the Fake news or mis-information. The dataset is divided into training set and testing set and decision tree algorithm is applied. The accuracy is calculated, actual and predicted values are compared. To help with the predictions you can draw a line, called a best-fit line that passes close to most of the data points. A heat map is a two-dimensional representation of data in which values are represented by colors. A heat map provides an immediate visual summary of information. More elaborate heat maps allow the viewer to understand complex data sets. Simple linear regression is an approach for predicting a response using a single feature. It is one of the most basic machine learning models that a machine learning enthusiast get to know about. It is assumed that the two variables are linearly related. Logistic regression is one of the most popular Machine Learning algorithms, which comes under the Supervised Learning technique. It is used for predicting the categorical dependent variable using a given set of independent variables. Random forest classifier is an ensemble tree-based machine learning algorithm. The random forest classifier is a set of decision trees from a randomly selected subset of the training set. Random forest is a Supervised Machine Learning Algorithm that is used widely in Classification and Regression problems. It builds decision trees on different samples.

3.2 Proposed System Diagram

The predictive model is used to prediction of the precipitation. The first step is converting data in to the correct format to conduct experiments then make a good analysis of data and observe variation in the patterns of rainfall. We predict the fake news by separating the dataset into training set and testing set then we apply different machine learning approaches an statistical techniques and compare and draw analysis over various approaches used. With the help of numerous approaches we attempt to minimize the error. Data analysis is done to achieve certainty of future result to be close so that prediction is valid and correctly interpreted. This certainty can

be gained only after raw data is verified and checked for abnormality thus ensuring that the data was gathered without any errors. It also helps in finding the data which contains irrelevant features for prediction model. Data pre-processing is a data mining technique that converts raw and inconsistent data into useful understandable format for the model. Raw data is inconsistent and incomplete and contains missing features along with many errors. We can also handle the missing values either by deleting irrelevant column or row. Encoding of categorical data is done as model is based on mathematical equations and calculations hence it is necessary to convert these categorical data into numeric.

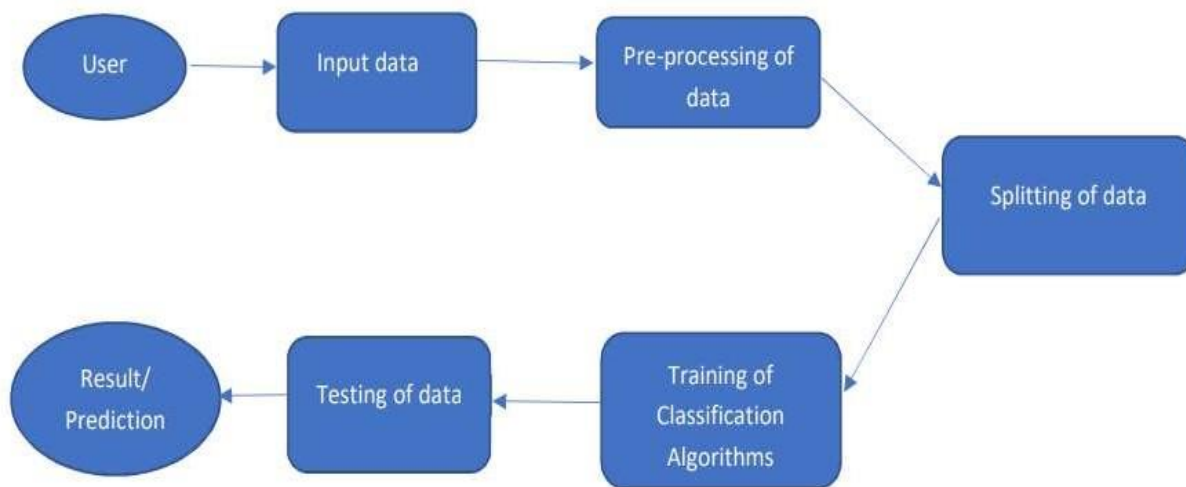


Figure 3.2.1 – Working of the model

As the dataset is very large, feature reduction is done so that it improves the accuracy, reduces the computation time and also storage. JupyterLab is the latest web-based interactive development environment for notebooks, code, and data. Its flexible interface allows users to configure and arrange workflows in data science, scientific computing, computational journalism, and machine learning.

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 Software Component

JUPYTERLAB

JupyterLab is the latest web-based interactive development environment for notebooks, code, and data. Its flexible interface allows users to configure and arrange workflows in data science, scientific computing, computational journalism, and machine learning. The Jupyter Notebook is the original web application for creating and sharing computational documents. It offers a simple, streamlined, document-centric experience.

KAGGLE

Kaggle is an online community platform for data scientists and machine learning enthusiasts. Kaggle allows users to collaborate with other users, find and publish datasets, use GPU integrated notebooks, and compete with other data scientists to solve data science challenges. Kaggle enables users to find and publish datasets, explore and build models in a web-based data science environment, work with other data scientists and machine learning engineers, and enter competitions to solve data science challenges.

CHAPTER 5

IMPORTED MODULES

5.1 PANDAS

Pandas is an open-source library that is made mainly for working with relational or labeled data both easily and intuitively. It provides various data structures and operations for manipulating numerical data and time series.

5.2 NUMPY

Numpy is a general-purpose array-processing package. It provides a high-performance multidimensional array object, and tools for working with these arrays. It is the fundamental package for scientific computing with Python.

5.3 SEABORN

Seaborn is a Python data visualization library based on matplotlib. It provides a high-level interface for drawing attractive and informative statistical graphics.

5.4 MATPLOTLIB

Matplotlib is easy to use and an amazing visualizing library in Python. It is built on NumPy arrays and designed to work with the broader SciPy stack and consists of several plots like line, bar, scatter, histogram, etc.

5.5 SKLEARN

Sklearn (scikit-learn) is a Python library that provides a wide range of unsupervised and supervised machine learning algorithms. It is also one of the most used machine learning libraries and is built on top of Scipy.

CHAPTER 6

ALGORITHMS USED

6.1 LOGISTIC REGRESSION

- **Linear Separability:** Logistic regression is a simple and effective algorithm for binary classification problems. If the fake news detection problem can be linearly separated into two classes (fake and real news), logistic regression can be a good choice.
- **Interpretability:** Logistic regression provides clear coefficients for each feature, making it easy to interpret the importance of different features in making classification decisions.

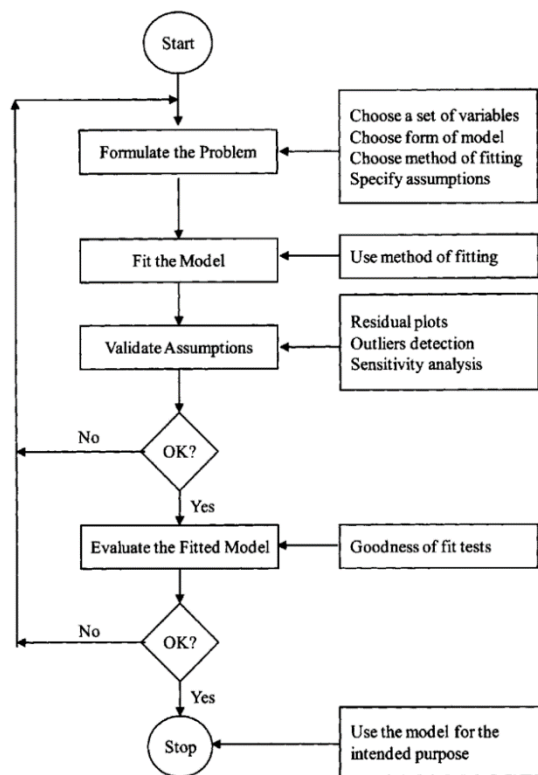


Fig 6.2.1 Logistic Regression Model

6.2 DECISION TREE

- **Non-linearity:** Decision trees can capture non-linear relationships in the data, which can be beneficial when the decision boundary between fake and real news is complex.
- **Interpretability:** Decision trees are interpretable, and the path of decisions in the tree can be analyzed to understand the decision-making process.

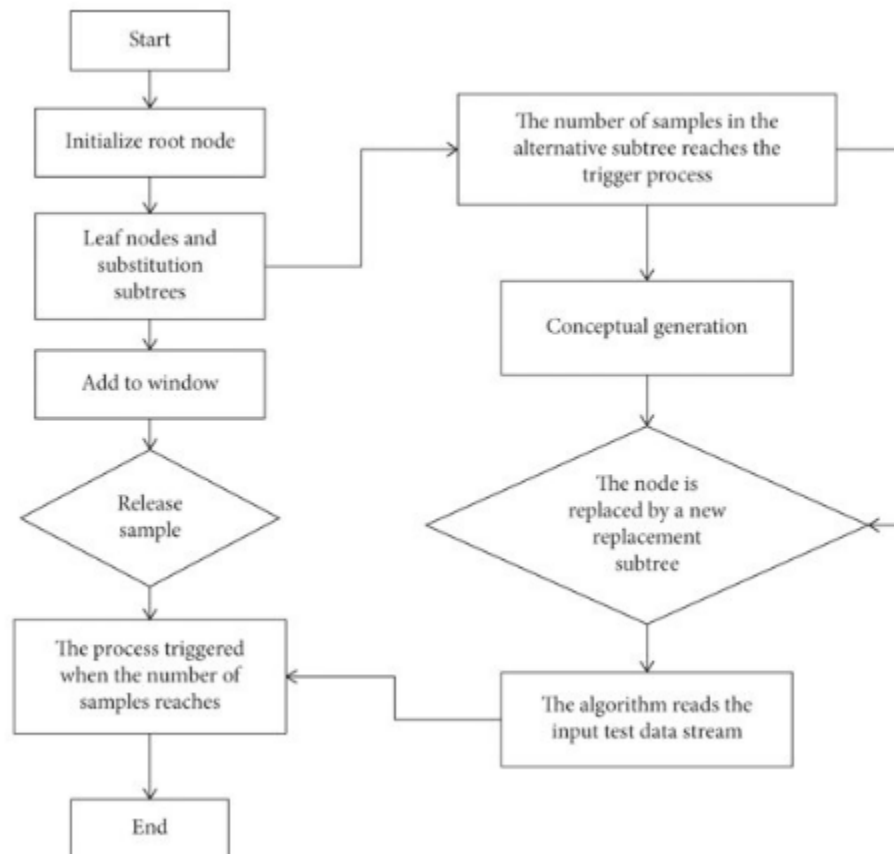


Fig 6.2.2 Decision Tree Model

6.3 GRADIENT BOOSTING CLASSIFIER

- **High Accuracy:** Gradient boosting methods, like XGBoost or LightGBM, often achieve high accuracy by combining the predictions of multiple weak classifiers (typically decision trees).

They are particularly effective when the data is imbalanced.

- **Robustness:** Gradient boosting algorithms are robust to outliers and can handle noisy data effectively.

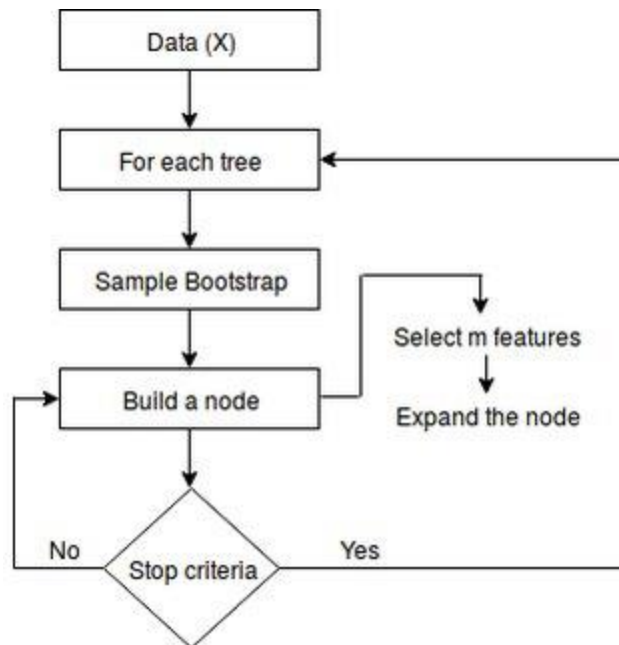


Fig 6.2.3 Gradient Boosting Model

6.4 RANDOM FOREST

- **Ensemble Learning:** Random forests are an ensemble of decision trees, which can provide improved generalization and reduce overfitting. They are a good choice when the data has many features.
- **Feature Importance:** Random forests can measure feature importance, helping to identify key features contributing to the classification.

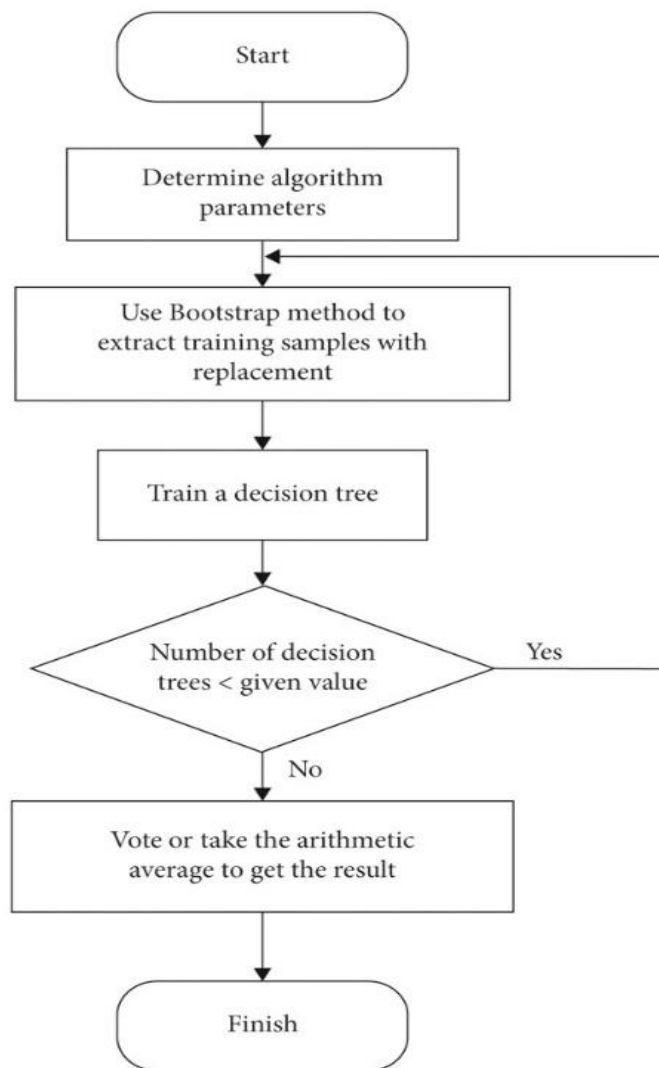


Fig 6.2.4 Random Forest Model

6.5 SVM

- **Effective in High-Dimensional Spaces:** SVMs are effective in high-dimensional feature spaces, making them suitable for text classification problems.
- **Maximal Margin:** SVM tries to find the maximum margin hyperplane, which can lead to better generalization and the ability to handle complex decision boundaries.

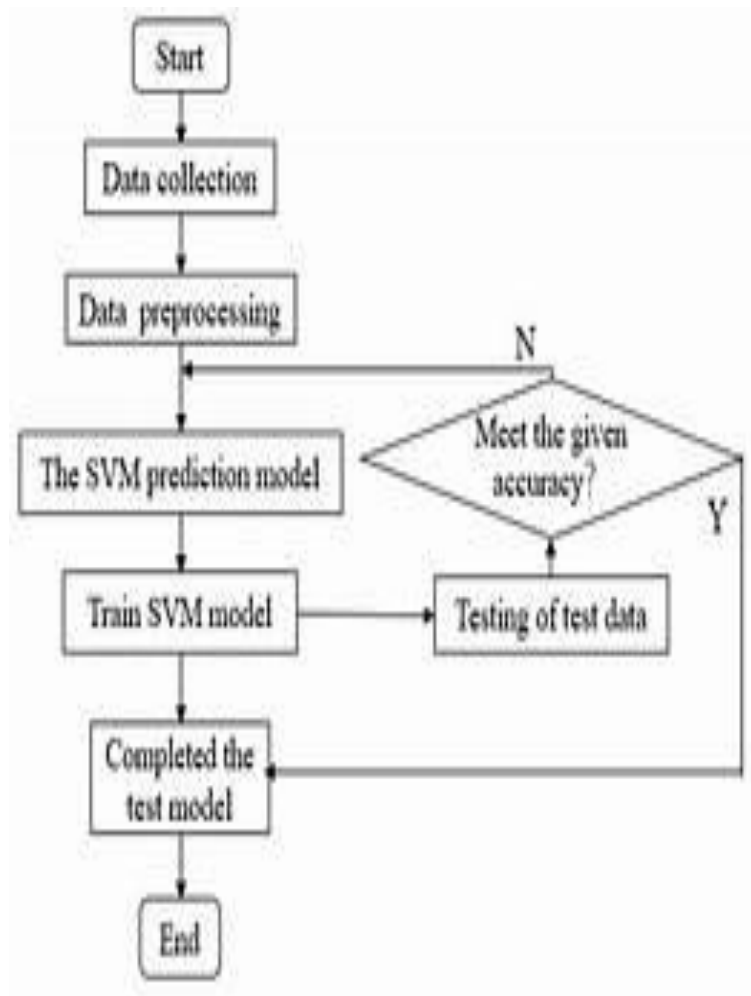


Fig 6.2.5 SVM Model

6.6 NAIVE BAYES CLASSIFIER

- **Text Classification:** Naive Bayes is commonly used in text classification tasks, including fake news detection. It's based on Bayes' theorem and is particularly useful for modeling the probability of observing a set of words given a class.
- **Efficiency:** Naive Bayes is computationally efficient and can handle large datasets with many features.

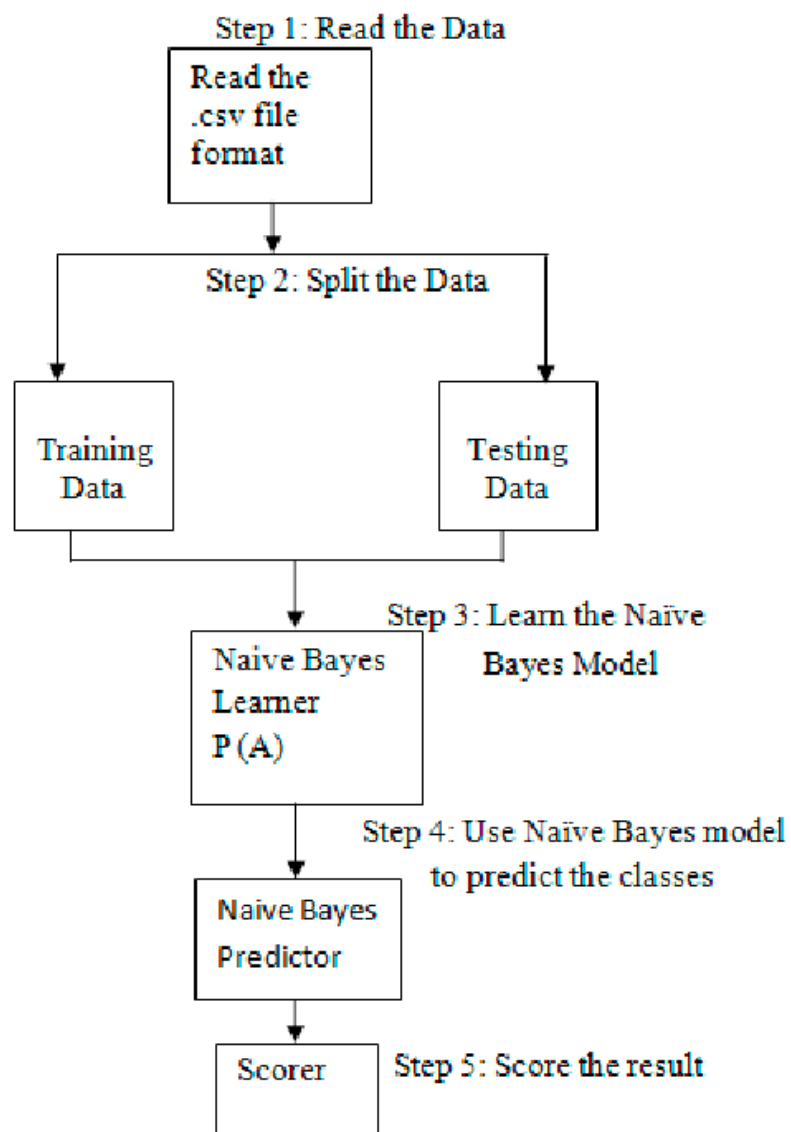


Fig 6.2.6 Naive Bayes Model

CHAPTER 7

IMPLEMENTATION

7.1 SOURCE CODE

Importing required library

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report
import re
import string
```

Inserting fake and real dataset

```
df_fake = pd.read_csv("Fake.csv")
df_true = pd.read_csv("True.csv")
df_true.head(10)
```

Inserting a column called "class" for fake and real news dataset to categories fake and true news.

```
df_fake["class"] = 0
df_true["class"] = 1
```

```
df_fake.shape, df_true.shape
```

BASIC DATA EXPLORATION

How many articles per subject?

```
import matplotlib.pyplot as plt
print(df_true.groupby(['subject'])['text'].count())
df_true.groupby(['subject'])['text'].count().plot(kind="bar")
plt.show()
```

How many articles per subject?

```
import matplotlib.pyplot as plt
print(df_fake.groupby(['subject'])['text'].count())
df_fake.groupby(['subject'])['text'].count().plot(kind="bar")
plt.show()
```

Word cloud for fake news

```

from wordcloud import WordCloud
fake_data = df_fake[df_fake["class"] == 0]
all_words = ' '.join([text for text in fake_data.text])
wordcloud = WordCloud(width= 800, height= 500,
                      max_font_size = 110,
                      collocations = False).generate(all_words)

plt.figure(figsize=(10,7))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis("off")
plt.show()

# Word cloud for real news
from wordcloud import WordCloud
real_data = df_true[df_true["class"] == 1]
all_words = ' '.join([text for text in fake_data.text])
wordcloud = WordCloud(width= 800, height= 500,
                      max_font_size = 110,
                      collocations = False).generate(all_words)

plt.figure(figsize=(10,7))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis("off")
plt.show()

import nltk
from nltk import tokenize
token_space = tokenize.WhitespaceTokenizer()
def counter(text, column_text, quantity):
    all_words = ' '.join([text for text in text[column_text]])
    token_phrase = token_space.tokenize(all_words)
    frequency = nltk.FreqDist(token_phrase)
    df_frequency = pd.DataFrame({"Word": list(frequency.keys()),
                                "Frequency":
list(frequency.values())})
    df_frequency = df_frequency.nlargest(columns = "Frequency", n =
quantity)
    plt.figure(figsize=(12,8))
    ax = sns.barplot(data = df_frequency, x = "Word", y = "Frequency",
color = 'blue')
    ax.set(ylabel = "Count")
    plt.xticks(rotation='vertical')
    plt.show()

# Most frequent words in fake news
import seaborn as sns
counter(df_fake[df_fake["class"] == 0], "text", 20)

```

```

counter(df_true[df_true["class"] == 1], "text", 20)
df_fake_manual_testing = df_fake.tail(10)
for i in range(23480,23470,-1):
    df_fake.drop([i], axis = 0, inplace = True)
df_true_manual_testing = df_true.tail(10)
for i in range(21416,21406,-1):
    df_true.drop([i], axis = 0, inplace = True)

```

Merging the manual testing dataframe in single dataset and save it in a csv file

```

df_manual_testing=pd.concat([df_fake_manual_testing,df_true_manual_testing], axis = 0)
df_manual_testing.to_csv("manual_testing.csv")

```

Merging the main fake and true dataframe

```

df_marge = pd.concat([df_fake, df_true], axis =0 )
df=df_marge.head(10)
df_marge.columns

```

#"title", "subject" and "date" columns is not required for detecting the fake news, so drop the columns.

```

df = df_marge.drop(["title", "subject","date"], axis = 1)
df.head(10)

```

Randomly shuffling the dataframe

```

df = df.sample(frac = 1)
df.head(10)
df.isnull().sum()
df.reset_index(inplace = True)
df.drop(["index"], axis = 1, inplace = True)
df.columns
df.head()

```

Creating a function to convert the text in lowercase, remove the extra space, special chr., ulr and links.

```

def wordopt(text):
    text = text.lower()
    text = re.sub('\[.*?\]', '', text)
    text = re.sub("\W", " ",text)
    text = re.sub('https?://\S+|www\.\S+', '', text)
    text = re.sub('<.*?>+', '', text)
    text = re.sub('[%s]' % re.escape(string.punctuation), '', text)
    text = re.sub('\n', '', text)
    text = re.sub('\w*\d\w*', '', text)
    return text

```

```
df["text"] = df["text"].apply(wordopt)
df.head(10)
```

Defining dependent and independent variable as x and y

```
x = df["text"]
y = df["class"]
```

Splitting the dataset into training set and testing set.

```
x_train, x_test, y_train, y_test = train_test_split(x, y,
test_size=0.25)
```

Convert text to vectors

```
from sklearn.feature_extraction.text import TfidfVectorizer
vectorization = TfidfVectorizer()
xv_train = vectorization.fit_transform(x_train)
xv_test = vectorization.transform(x_test)
```

1. Logistic Regression

```
from sklearn.linear_model import LogisticRegression
LR = LogisticRegression()
LR.fit(xv_train, y_train)
LR.score(xv_test, y_test)
pred_lr=LR.predict(xv_test)
print(classification_report(y_test, pred_lr))
from sklearn import metrics
import itertools
def plot_confusion_matrix(cm, classes,
                          normalize=False,
                          title='Confusion matrix',
                          cmap=plt.cm.Blues):

    plt.imshow(cm, interpolation='nearest', cmap=cmap)
    plt.title(title)
    plt.colorbar()
    tick_marks = np.arange(len(classes))
    plt.xticks(tick_marks, classes, rotation=45)
    plt.yticks(tick_marks, classes)
    if normalize:
        cm = cm.astype('float') / cm.sum(axis=1)[:, np.newaxis]
        print("Normalized confusion matrix")
    else:
        print('Confusion matrix, without normalization')

    thresh = cm.max() / 2.
    for i, j in itertools.product(range(cm.shape[0]),
```

```

range(cm.shape[1])):
    plt.text(j, i, cm[i, j],
             horizontalalignment="center",
             color="white" if cm[i, j] > thresh else "black")
plt.tight_layout()
plt.ylabel('True label')
plt.xlabel('Predicted label')
from sklearn import metrics
cm = metrics.confusion_matrix(y_test, pred_lr)
plot_confusion_matrix(cm, classes=['Fake', 'Real'])
Confusion matrix, without normalization

```

DECISION TREE CLASSIFIER

```

from sklearn.tree import DecisionTreeClassifier

```

```

DT = DecisionTreeClassifier()
DT.fit(xv_train, y_train)

```

```

pred_dt = DT.predict(xv_test)
DT.score(xv_test, y_test)
print(classification_report(y_test, pred_dt))

```

```

cm = metrics.confusion_matrix(y_test, pred_dt)
plot_confusion_matrix(cm, classes=['Fake', 'Real'])
Confusion matrix, without normalization

```

GRADIENT BOOSTING CLASSIFIERS

```

from sklearn.ensemble import GradientBoostingClassifier
GB=GradientBoostingClassifier(random_state=0)
GB.fit(xv_train, y_train)

```

```

pred_gb = GB.predict(xv_test)
GB.score(xv_test, y_test)
print(classification_report(y_test, pred_gb))

```

```

cm = metrics.confusion_matrix(y_test, pred_gb)
plot_confusion_matrix(cm, classes=['Fake', 'Real'])
Confusion matrix, without normalization

```

RANDOM FOREST CLASSIFIER

```

from sklearn.ensemble import RandomForestClassifier

```

```

RF = RandomForestClassifier(random_state=0)

```

```
RF.fit(xv_train, y_train)
```

```
pred_rf = RF.predict(xv_test)
RF.score(xv_test, y_test)
print(classification_report(y_test, pred_rf))
```

```
cm = metrics.confusion_matrix(y_test, pred_rf)
plot_confusion_matrix(cm, classes=['Fake', 'Real'])
Confusion matrix, without normalization
```

SVM

```
from sklearn import svm
```

```
#Create a svm Classifier
SVM = svm.SVC(kernel='linear') # Linear Kernel
SVM.fit(xv_train, y_train)
```

```
pred_svm = SVM.predict(xv_test)
SVM.score(xv_test, y_test)
print(classification_report(y_test, pred_svm))
```

```
cm = metrics.confusion_matrix(y_test, pred_svm)
plot_confusion_matrix(cm, classes=['Fake', 'Real'])
Confusion matrix, without normalization
```

NAIVE BAYES

```
from sklearn.naive_bayes import MultinomialNB
```

```
NB = MultinomialNB()
NB.fit(xv_train, y_train)
```

```
pred_nb=NB.predict(xv_test)
NB.score(xv_test, y_test)
print(classification_report(y_test, pred_nb))
cm = metrics.confusion_matrix(y_test, pred_nb)
plot_confusion_matrix(cm, classes=['Fake', 'Real'])
Confusion matrix, without normalization
```

```
from sklearn.metrics import accuracy_score
```

```
dct = dict()
dct['LogisticRegression'] = round(accuracy_score(y_test, pred_lr)*100,2)
dct['DecisionTreeClassifier'] = round(accuracy_score(y_test,
```

```

pred_dt)*100,2)
dct[' GradientBoosting'] = round(accuracy_score(y_test, pred_gb)*100,2)
dct['RandomForest'] = round(accuracy_score(y_test, pred_rf)*100,2)
dct[' SVM'] = round(accuracy_score(y_test, pred_svm)*100,2)
dct['Naive Bayes'] = round(accuracy_score(y_test, pred_nb)*100,2)
import matplotlib.pyplot as plt
plt.figure(figsize=(8,7))
plt.bar(list(dct.keys()),list(dct.values()))
plt.ylim(90,100)
plt.yticks((91, 92, 93, 94, 95, 96, 97, 98, 99, 100))

import pickle
filename = 'finalised_model.pkl'
pickle.dump(LR,open(filename,'wb'))
filename = 'vectorizer.pkl'
pickle.dump(vectorization,open(filename,'wb'))
Model Testing With Manual Entry
News
def output_lable(n):
    if n==0:
        return "Fake News"
    elif n==1:
        return "True News"
def manual_testing(news):
    testing_news = {"text":[news]}
    new_def_test = pd.DataFrame(testing_news)
    new_def_test["text"] = new_def_test["text"].apply(wordopt)
    new_x_test = new_def_test["text"]
    new_xv_test = vectorization.transform(new_x_test)
    pred_LR = LR.predict(new_xv_test)
    pred_DT = DT.predict(new_xv_test)
    pred_GB = GB.predict(new_xv_test)
    pred_RF = RF.predict(new_xv_test)
    return print("\n\nLR Prediction: {} \nDT Prediction: {} \nGB
Prediction: {} \nRF Prediction:
{}".format(output_lable(pred_LR[0]),output_lable(pred_DT[0]),output_labl
e(pred_GB[0]),output_lable(pred_RF[0])))

    #return print("\n\nLR Prediction:
{}".format(output_lable(pred_LR[0])))
news = str(input())
manual_testing(news)

```

7.2 OUTPUT

	title	text	subject	date
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...	politicsNews	December 31, 2017
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...	politicsNews	December 29, 2017
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...	politicsNews	December 31, 2017
3	FBI Russia probe helped by Australian diplomat...	WASHINGTON (Reuters) - Trump campaign adviser ...	politicsNews	December 30, 2017
4	Trump wants Postal Service to charge 'much mor...	SEATTLE/WASHINGTON (Reuters) - President Donal...	politicsNews	December 29, 2017
5	White House, Congress prepare for talks on spe...	WEST PALM BEACH, Fla./WASHINGTON (Reuters) - T...	politicsNews	December 29, 2017
6	Trump says Russia probe will be fair, but time...	WEST PALM BEACH, Fla (Reuters) - President Don...	politicsNews	December 29, 2017
7	Factbox: Trump on Twitter (Dec 29) - Approval ...	The following statements were posted to the ve...	politicsNews	December 29, 2017
8	Trump on Twitter (Dec 28) - Global Warming	The following statements were posted to the ve...	politicsNews	December 29, 2017
9	Alabama official to certify Senator-elect Jone...	WASHINGTON (Reuters) - Alabama Secretary of St...	politicsNews	December 28, 2017

Fig 7.2.1 Dataset

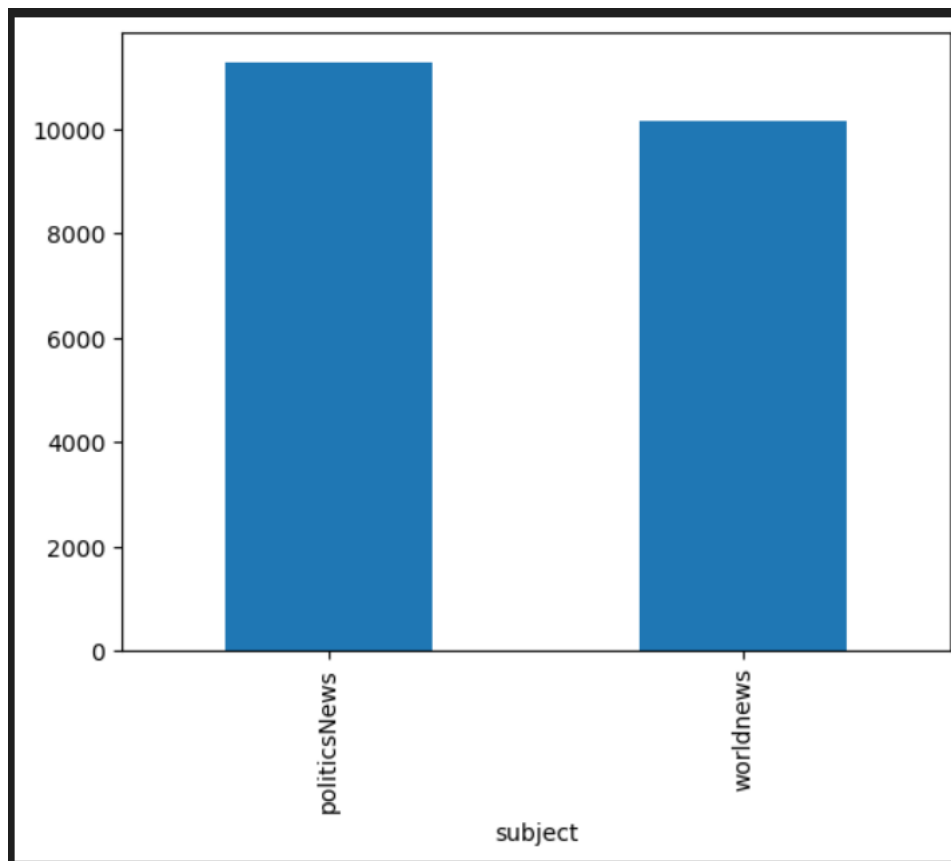


Fig 7.2.2 Article types in true dataset

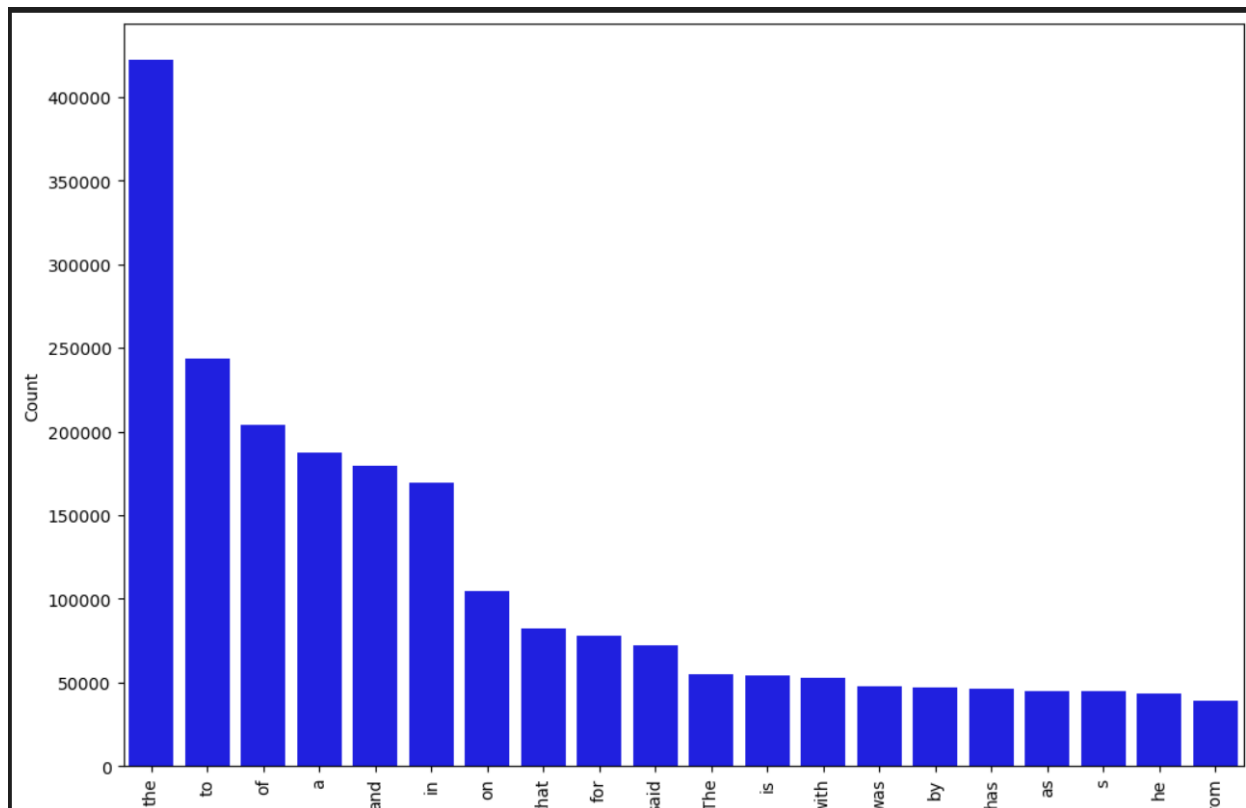


Fig 7.2.7 Frequent words in true news

	text	class
0	Donald Trump just couldn t wish all Americans ...	0
1	House Intelligence Committee Chairman Devin Nu...	0
2	On Friday, it was revealed that former Milwauk...	0
3	On Christmas day, Donald Trump announced that ...	0
4	Pope Francis used his annual Christmas Day mes...	0
5	The number of cases of cops brutalizing and ki...	0
6	Donald Trump spent a good portion of his day a...	0
7	In the wake of yet another court decision that...	0
8	Many people have raised the alarm regarding th...	0
9	Just when you might have thought we d get a br...	0

Fig 7.2.8 After removing title, subject, date

	text	class
17094	CARACAS/PUERTO ORDAZ, Venezuela (Reuters) - Ve...	1
11236	WASHINGTON (Reuters) - Republican presidential...	1
251	WASHINGTON (Reuters) - Millions of households ...	1
6224	Some of Donald Trump s biggest fans and suppor...	0
15045	Hillary Clinton is a serial liar. Hillary Cli...	0
19103	LONDON (Reuters) - Irish Prime Minister Leo Va...	1
13992	ANKARA (Reuters) - Turkish authorities issued ...	1
1076	It s hard not to feel that John McCain, after ...	0
19567	MOSCOW (Reuters) - At least eight North Korean...	1
8934	On Tuesday s edition of The Nightly Show, Dem...	0

Fig 7.2.9 Randomly Shuffling Dataframe

	text	class
0	caracas puerto ordaz venezuela reuters ve...	1
1	washington reuters republican presidential...	1
2	washington reuters millions of households ...	1
3	some of donald trump s biggest fans and suppor...	0
4	hillary clinton is a serial liar hillary cli...	0
5	london reuters irish prime minister leo va...	1
6	ankara reuters turkish authorities issued ...	1
7	it s hard not to feel that john mccain after ...	0
8	moscow reuters at least eight north korean...	1
9	on tuesday s edition of the nightly show dem...	0

Fig 7.2.10 Conversion into lowercase

	precision	recall	f1-score	support
0	0.99	0.99	0.99	5858
1	0.99	0.99	0.99	5362
accuracy			0.99	11220
macro avg	0.99	0.99	0.99	11220
weighted avg	0.99	0.99	0.99	11220

Fig 7.2.11 Logistic Regression report

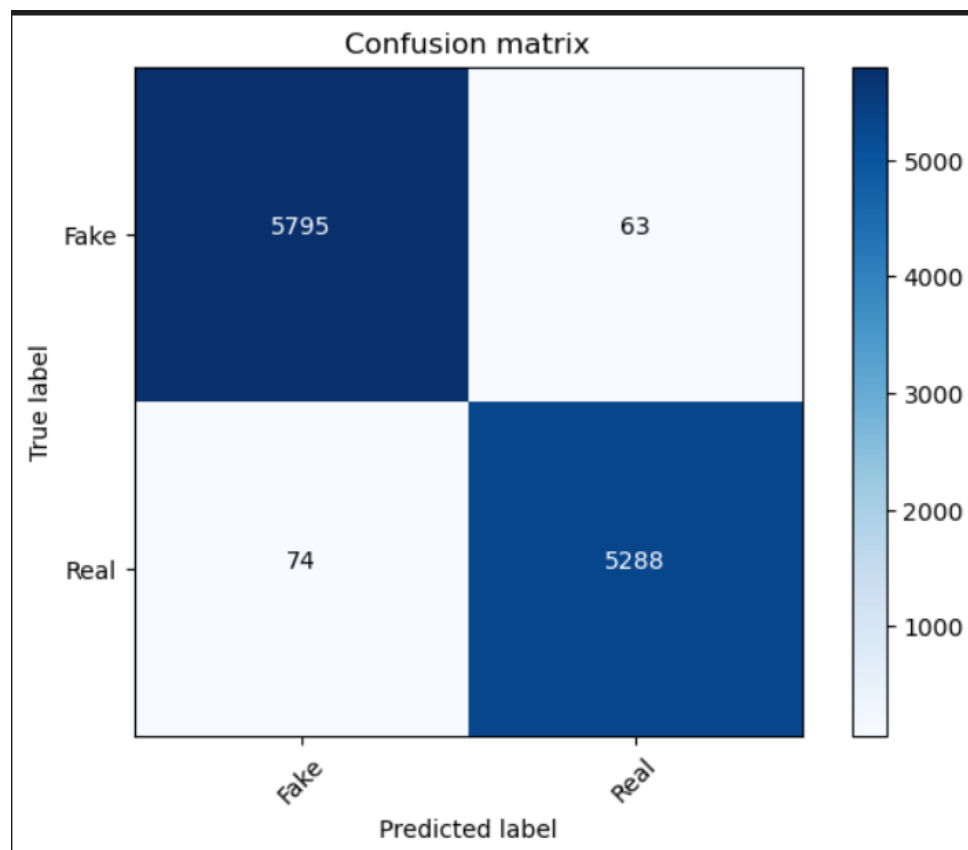


Fig 7.2.12 Linear Regression matrix

	precision	recall	f1-score	support
0	0.99	1.00	1.00	5858
1	1.00	0.99	1.00	5362
accuracy			1.00	11220
macro avg	1.00	1.00	1.00	11220
weighted avg	1.00	1.00	1.00	11220

Fig 7.2.13 Decision Tree Report

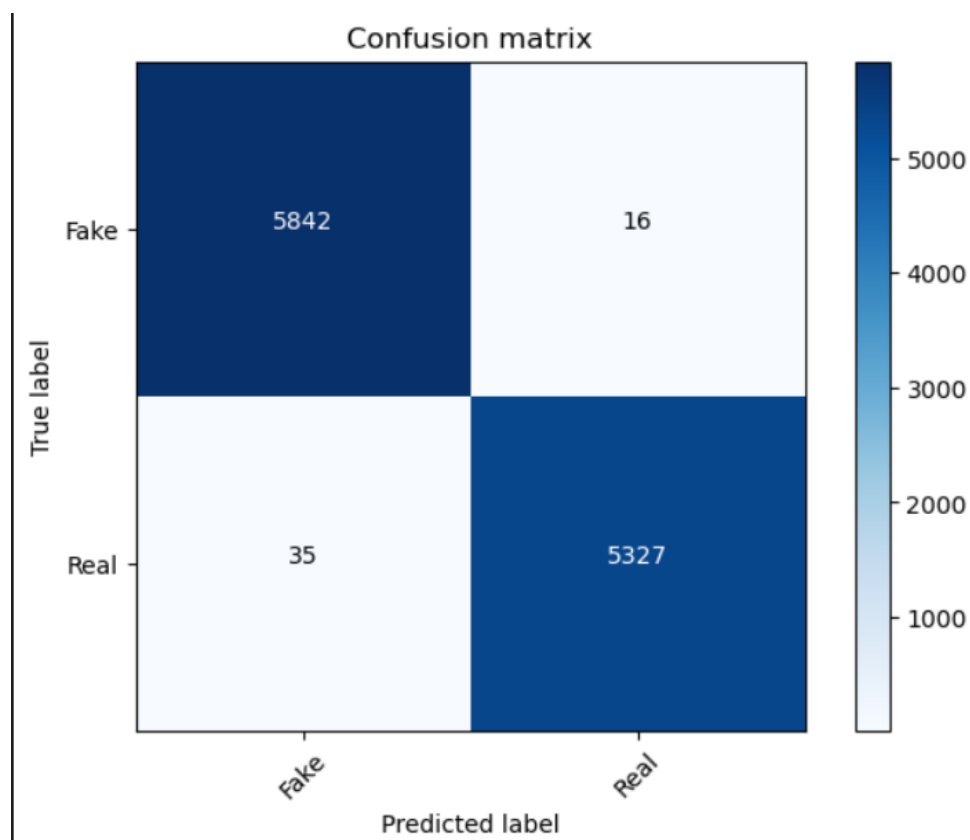


Fig 7.2.14 Decision Tree Matrix

	precision	recall	f1-score	support
0	1.00	0.99	1.00	5858
1	0.99	1.00	1.00	5362
accuracy			1.00	11220
macro avg	1.00	1.00	1.00	11220
weighted avg	1.00	1.00	1.00	11220

Fig 7.2.15 Gradient Boosting report

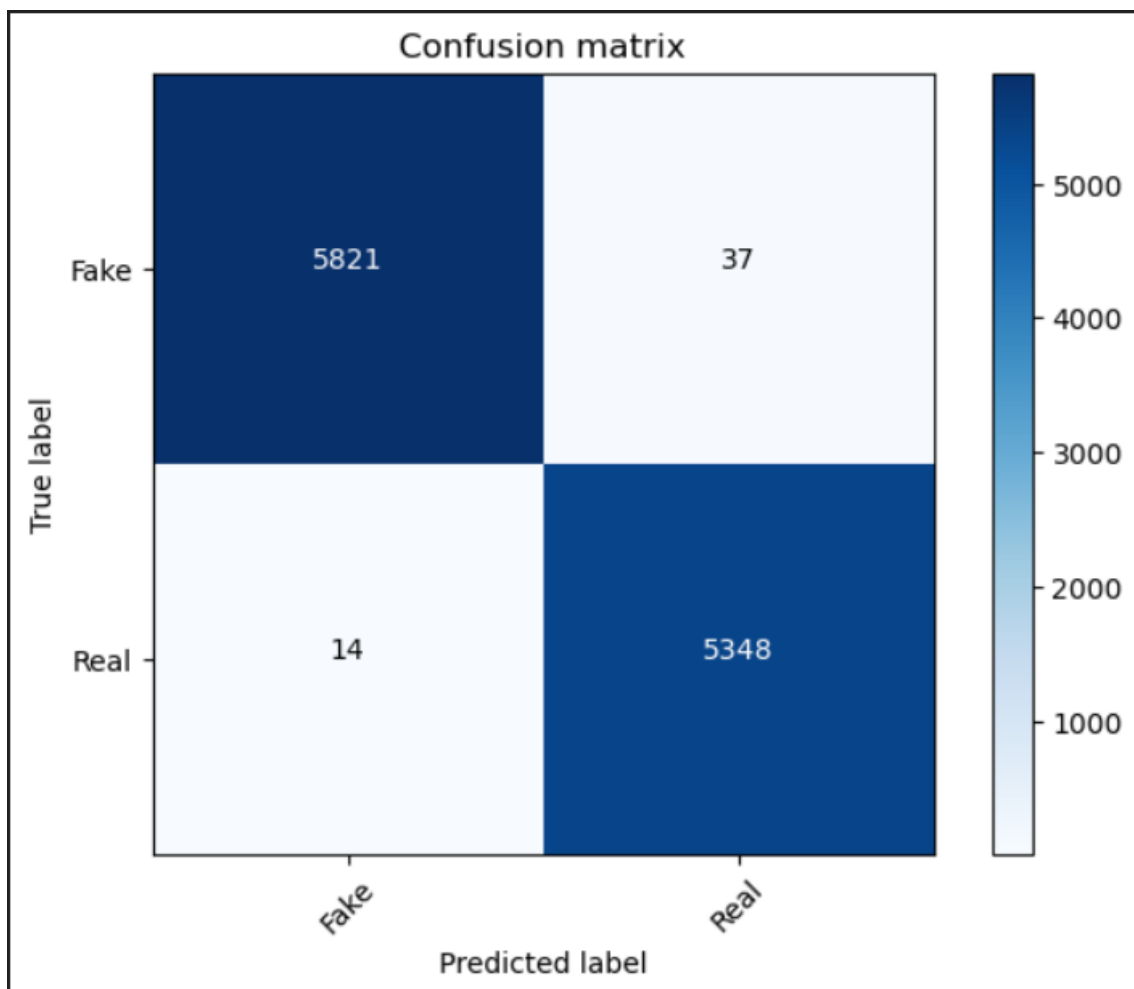


Fig 7.2.16 Gradient Boosting Matrix

	precision	recall	f1-score	support
0	0.99	0.99	0.99	5858
1	0.99	0.99	0.99	5362
accuracy			0.99	11220
macro avg	0.99	0.99	0.99	11220
weighted avg	0.99	0.99	0.99	11220

Fig 7.2.17 Random Forest report

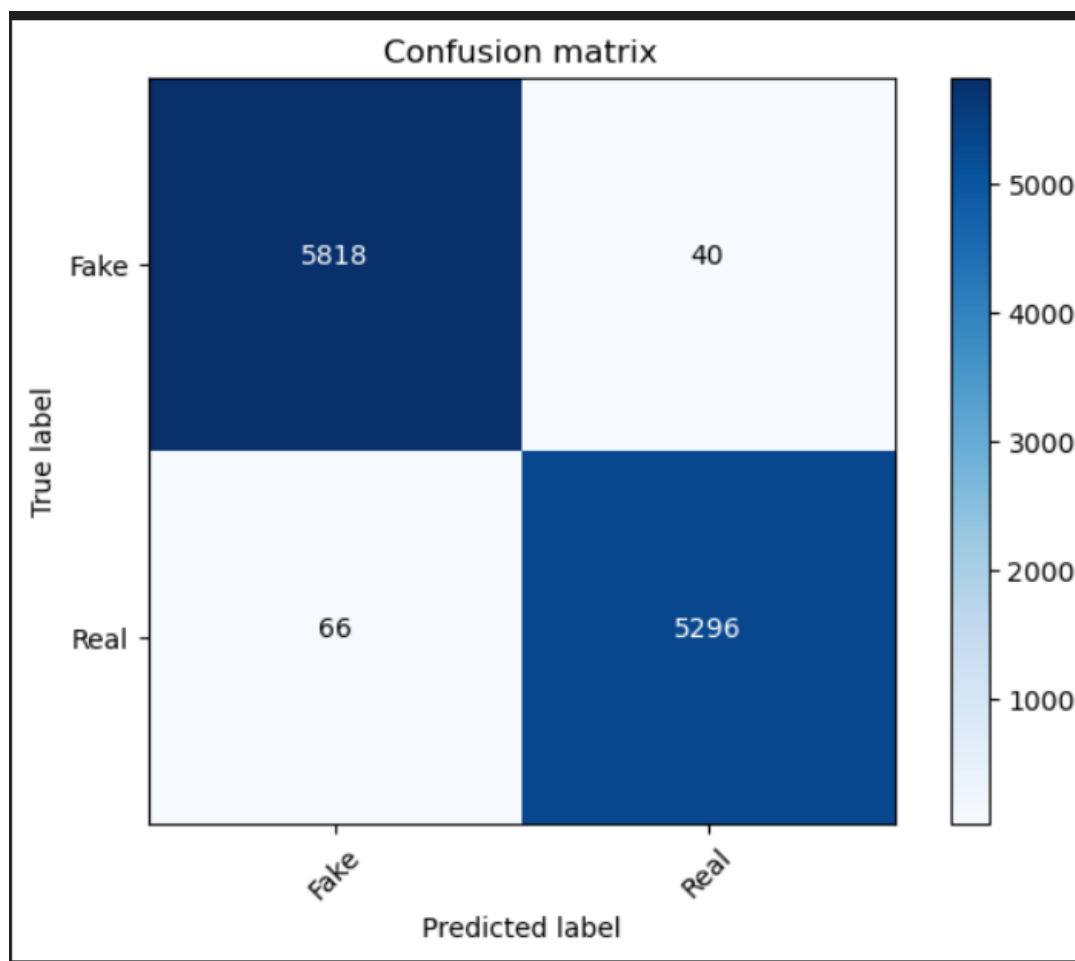


Fig 7.2.18 Random Forest report

	precision	recall	f1-score	support
0	0.94	0.95	0.94	5889
1	0.94	0.93	0.94	5331
accuracy			0.94	11220
macro avg	0.94	0.94	0.94	11220
weighted avg	0.94	0.94	0.94	11220

Fig 7.2.19 Naive Bayes Report

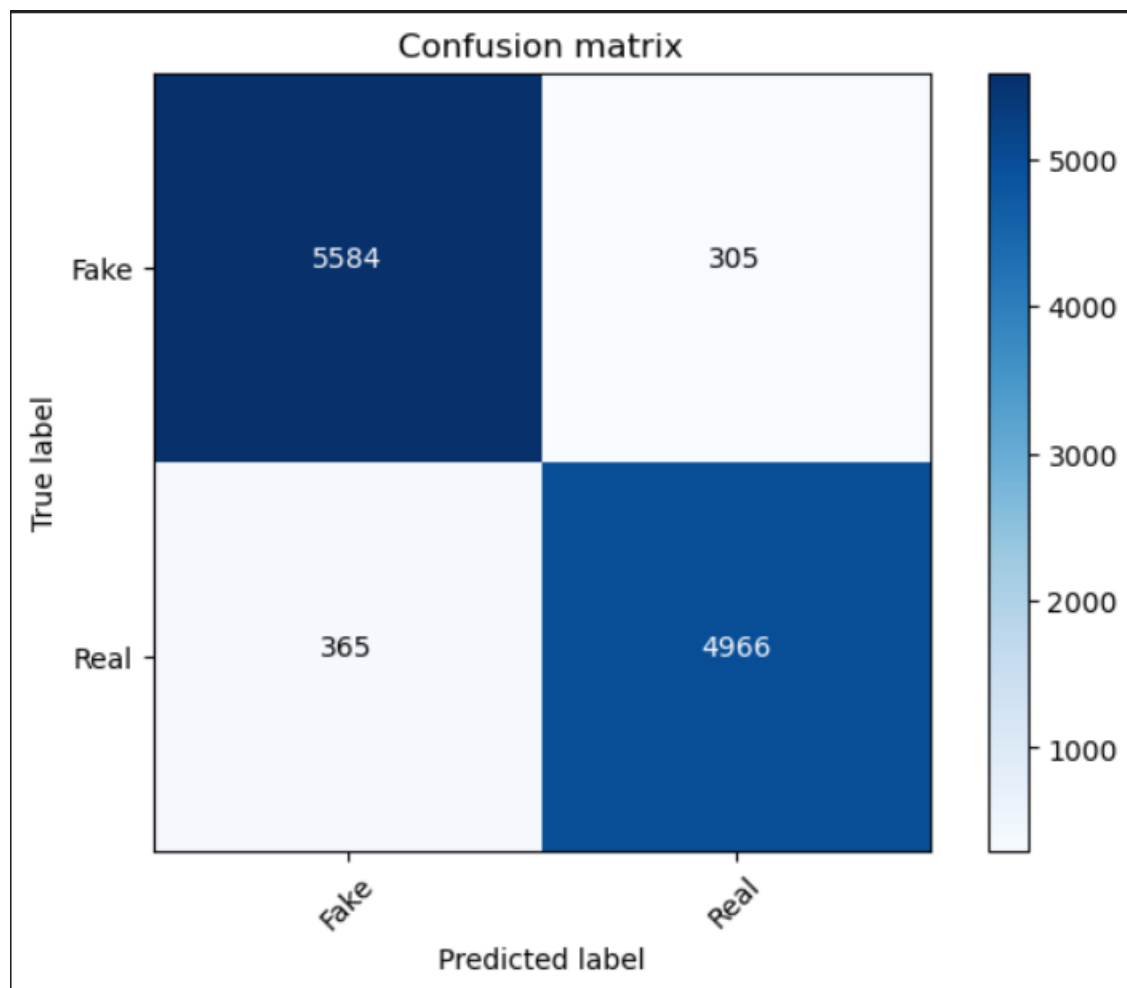


Fig 7.2.20 Naïve Bayes matrix

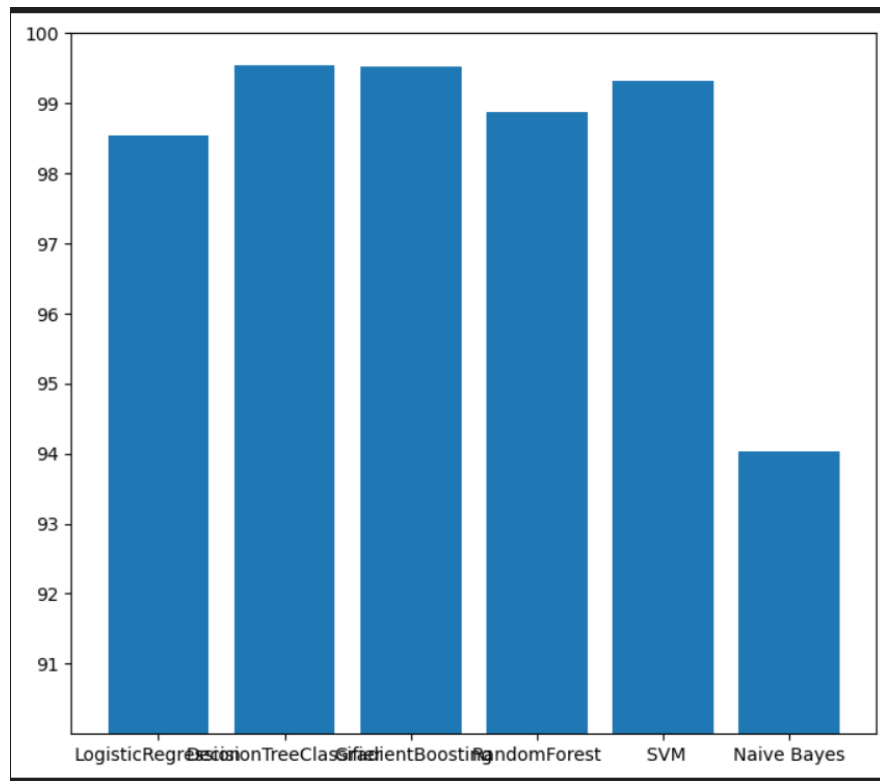


Fig 7.2.21 Accuracy of algorithms

Donald Trump just couldn't wish all Americans a Happy New Year and leave it at that. Instead, he had to give a shout out

LR Prediction: Fake News

DT Prediction: Fake News

GB Prediction: Fake News

RF Prediction: Fake News

Fig 7.2.22 Prediction of news

CHAPTER 8

CONCLUSION:

Our project has demonstrated that machine learning models, when trained on large and diverse datasets, can achieve high accuracy in classifying news articles as either genuine or fake. This represents a promising step towards combating the spread of false information and protecting the integrity of news and information sources. The creators of fake news are constantly adapting their strategies, making it crucial for our detection systems to remain dynamic and up-to-date. Future work in this field should focus on continuous model improvement, the exploration of more sophisticated features, and an emphasis on collaborative efforts to counteract misinformation on a broader scale. We hope that our project has shown promise, no algorithm can be 100% foolproof. Human fact-checkers and critical thinking will remain essential in the fight against fake news.

REFERENCES

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<https://matplotlib.org>